5. Practice Session - II

Friday, 8 November 2024 8:37 PM

(1.) Are these system specifications consistent?

A: Whenever the system software is being upgraded, users cannot access the file system.

p

q r

B: If users can access the file system, then they can save new files.

C: If users cannot save new files, then the system software is not being upgraded.

Sol.

$$
\left.
\begin{aligned}
A &: \; p \rightarrow \neg q \\
B &: \; q \rightarrow r \\
C &: \; \neg r \rightarrow \neg p
\end{aligned}
\right\}
$$

(A) (B) (C)

$C : \neg r \rightarrow \neg p$

p	q	r	$\neg p$	$\neg q$	$\neg r$	A: $p \rightarrow \neg q$	B: $q \rightarrow r$	C: $\neg r \rightarrow \neg p$	
T	T	T	F	F	F	F ✗	—	—	
T	T	F	F	F	T	F ✗	—	—	
$\boxed{T}$	$\boxed{F}$	$\boxed{T}$	F	T	F	T	T	T	∴
T	F	F	F	T	T	T	T		
F	T	T	T	F	F	T	T		
F	T	F	T	F	T	T	F ✗	—	
F	F	T	T	T	F	T	T		
F	F	F	T	T	T	T	T		

argument form :

P_1

P_2

P_3

$\vdots$

P_n

$\boxed{\therefore\ Q}$

Yes/No ?

$(P_1 \wedge P_2 \ldots \wedge P_n) \rightarrow Q$ → it should be a tautology.

→ Sound valid arguments.

→ unsound ones. $\boxed{F \rightarrow F/T}$ ⇒ Valid.

Valid arguments are always consistent. ✗

$A \rightarrow B$ → true.

F T/F

(2.) A: If the file system is not locked, then new messages will be queued.

(H.W.)

B: If the file system is not locked, then the system is functioning normally, and conversely.

C: If new messages are not queued, then they will be sent to the message buffer.

D: If the file system is not locked, then new messages will be sent to the message buffer.

E: New messages will not be sent to the message buffer.

Consistent or not?

③ Are they logically equivalent :

a.) $(p \to q) \to r$ and $p \to (q \to r)$ ✗

(H.W.) b.) $(p \wedge q) \to r$ and $(p \to r) \wedge (q \to r)$ ✗

Sol.

p	q	r	$p \to q$ (x)	$(q \to r)$ (y)	$x \to r$	$p \to y$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	F	T	F	$\boxed{F}$	$\boxed{T}$
F	F	T	T	T	T	
F	F	F	T	T	F	

✗ not equivalent.

(4.) Find the dual of each of these compound propositions.

a) $\boxed{p \vee \neg p} = S \Rightarrow \text{Dual}(S) = S^*$

b) $p \wedge (q \vee (r \wedge T))$

c) $(p \wedge \neg q) \vee (q \wedge F)$

Any compound proposition that contains $\wedge$, $\vee$ and $\neg$, then its dual will be:

Just replace them

$$\begin{cases} \wedge \longrightarrow \vee \\ \vee \longrightarrow \wedge \\ T \longrightarrow F \\ F \longrightarrow T \end{cases}$$

Sol.

a) dual of $(p \vee \neg p) \Rightarrow p \wedge \neg p$

b) $p \vee (q \wedge (r \vee F))$

c) $(p \vee \neg q) \wedge (q \vee T)$

Q. Does $S^* = S$? no.

* Does $(S^*)^* = S$? Yes., $S \to$ compound proposition.

(5.) Determine whether these arguments are valid or not.

a) Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore Mia is not enrolled in the university.

Valid.

modus tollens.

b) Quincy likes all action movies. Quincy likes the movie Eight Men Out. Therefore, Eight Men Out is an action movie.

Invalid.

c) A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive.

Invalid.

fallacies.

(6.) a) If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number.

Invalid.

p (above "$x^2 \neq 0$"), q (above "$x \neq 0$")

b) If $x^2 \neq 0$, where x is a real number, then $x \neq 0$. Let a be a real number with $a^2 \neq 0$; then $a \neq 0$. $\therefore$

Valid.

Determine whether these arguments are valid or not.

Sol. a) $a = -5$, $\boxed{a^2 = 25}$ $\therefore$

b)

$$\begin{array}{l} p \to q \\ p \\ \hline \therefore q. \end{array}$$

modus ponens.